

SOURAV KUMAR VERMA

Associate Lead Engineer

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CAREER SUMMARY

A results-driven software developer with 7 plus years of experience in CyberSecurity and Automotive BSW software development with leading Tier 1 suppliers for major OEMs including Ford, General Motors, and Holon. Proven expertise in leading international software development teams and delivering high-quality software. Proficient in C, C++ programming and AutoSAR. Strong foundation in Electronics and Communications Engineering with a Bachelor of Technology degree.

TECHNICAL SKILLS

C | C++ | Hardware Security Module (HSM) | Secure Boot | Secure Access | Public Key Infrastructure | Key Management | Secure On Board Communication (SecOC) | JTAG Lockout | UDS | ISO 14229 | CAN | Diagnostics Stack | Crypto Stack | Hardware In Loop HIL Testing | AutoSAR | ASPICE | Microcontrollers | Renesas | Problem Solving | Encryption | AES | RSA | RTOS | Embedded Systems

TOOLS

Vector Da Vinci | CANoe | CANalyser | Green Hills Multi | Eclipse | IBM Rational DOORS | ETAS LabCar | Candela Studio | GenY | Enterprise Architect | HP Quality Center | IBM Jazz | Jenkins

PROFESSIONAL EXPERIENCE

ASSOCIATE LEAD ENGINEER | Nexteer | Bengaluru, India

Jan 2023 – present

Led a team of 8 software engineers, driving daily stand-up meetings and ensuring the timely delivery of high-quality software, with ASPICE and V Model compliance. Worked on software development, requirements engineering, and testing.

- Developed and implemented the Secure Boot feature to ensure the integrity and authenticity of the system during startup, protecting against unauthorized code execution.
- Designed and integrated cryptographic Key Provisioning and Management features to securely handle keys, ensuring proper key lifecycle management throughout the vehicle system.
- Led the development of Secure Access features to restrict unauthorized access to sensitive system data and configurations, enhancing overall system security.
- Implemented JTAG Lockout functionality to prevent unauthorized external access to debugging interfaces, ensuring the security of embedded systems.
- Developed and deployed Secure On-Board Communication protocols using Hardware Security Module (HSM) features, strengthening encryption and secure data transmission between system components.

SENIOR SOFTWARE ENGINEER | Bosch | Bengaluru, India

Sept 2018 – Jan 2023

Worked on software development, requirements engineering, and testing.

- Developed the Encrypted Passive Anti-Theft System (EPATS) to enhance ECU security by implementing a seed-key mechanism for authentication. Utilized AES encryption and decryption to restrict access to safety-critical data and configurations, ensuring secure access through successful authentication.
- Developed and integrated PARSED (Processing and Reporting System for Efficient Data), a unified system for accessing and reporting ECU data to the OEM's Cloud via OVTP communication channels. Involved in integrating the device driver from Vector into the ABS ECU and customizing the system for specific customer requirements.
- Developed and managed the complete Software Development Life Cycle (SDLC) for Unified Diagnostic Service (UDS) based on ISO 14229 for ABS ECUs, including development and testing on Renesas-based ECUs.

EDUCATION

BACHELOR'S DEGREE | B. Tech in Electronics and Communication | C V Raman College of Engineering, India

2014 – 2018

LANGUAGES

English - Cambridge University Certified Business English